

Frida – Introduction to Dynamic Analysis

Frida ek powerful dynamic instrumentation toolkit hai jo aapko running applications ke behavior ko real-time mein analyze, modify, aur trace karne ki facility deta hai . Isko "Greasemonkey for native apps" bhi kaha jata hai kyunke yeh aapko JavaScript snippets inject karne deta hai .

Frida Kya Hai?

Frida ek **dynamic code instrumentation toolkit** hai jo aapko running processes mein JavaScript code inject karne ki suvidha deta hai . Iska matlab hai ke aap app ko recompile kiye baghair, uske behavior ko modify kar sakte hain .

- Android, iOS, Windows, macOS, Linux, tvOS, watchOS, aur FreeBSD par kaam karta hai
- Java (Android) aur Objective-C (iOS) methods ko hook kar sakta hai
- Native functions ko bhi intercept kar sakta hai

🌸 **Frida Kaise Kaam Karti Hai?**

Frida target process ke memory mein **QuickJS JavaScript engine** inject karti hai . Yeh engine process ke andar run hota hai aur memory, functions, aur objects ko access kar sakta hai .

Basic Architecture:

1. **PTrace Hijack** – Frida ptrace use karke running process ke ek thread ko hijack karta hai
2. **Memory Allocation** – Ek chunk allocate karta hai aur bootstrapper populate karta hai
3. **Agent Injection** – frida-agent.so load karta hai
4. **Bi-Directional Communication** – Agent aur tool ke beech communication channel establish hota hai
5. **Thread Resume** – Hijacked thread restore ho jati hai

🚀 **Main Features**

Interceptor API – Functions ke entry aur exit par custom code run karne ke liye **Stalker API** – High-performance code tracing aur JIT recompilation ke liye **Java API (Android)** – Classes, methods, objects, aur heap scanning ke liye **ObjC API (iOS)** – Runtime classes, methods, aur live instances ke liye

🔧 **Frida Tools**

Frida install karne ke baad yeh tools available ho jate hain:

frida – CLI tool hai interactive aur script-based hooking ke liye **frida-trace** – Function calls ko dynamically trace karta hai **frida-ps** – Running processes aur installed apps list karta hai **frida-ls-devices** – Connected devices list karta hai

Android Setup

Step 1: frida-server download karein

Frida releases page se frida-server download karein

Step 2: Device par copy karein

```
adb push frida-server /data/local/tmp/  
adb shell "chmod 755 /data/local/tmp/frida-server"
```

Step 3: Run karein

```
adb shell "su -c /data/local/tmp/frida-server &"
```

Step 4: Verify karein

```
frida-ps -U
```

Simple Script Example

Frida script jo Android app ki method ko hook karta hai :

```
Java.perform(function () {  
    var Activity =  
Java.use("android.app.Activity");  
    Activity.onResume.implementation = function ()  
{  
        console.log("[*] onResume() got called!");  
        this.onResume();  
    };  
});
```

Modes of Operation

Injected Mode – frida-server running hai root/jailbreak device par **Embedded Mode** – frida-gadget library embedded hai app mein **Preloaded Mode** – frida-gadget autonomously script load karta hai

Real-World Examples

Root Detection Bypass: isDeviceRooted() method ko override karke false return karaya **In-App Score**

Modification: Encryption method ko hook karke score value change ki **Flag Validation Bypass:** frida-trace se method identified aur return value modified

 **Educational Purpose Only:** Frida ka istemal sirf apne own apps aur authorized testing environments mein karein. Kisi doosre ke app mein bina permission ke dynamic analysis karna applicable cyber laws ke under punishable hai .

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HAPPY LEARNING! STAY ETHICAL
